Project Proposal

Part 1 of 9 • Group Project Assignment

1 Project Title

Gym Membership Management System

2 Problem Description

Many small and medium-sized gyms still record members, subscriptions, and payments on paper or in scattered spreadsheets. This makes it hard to know who is an active member, when a subscription expires, and who has not paid.

The problem affects **gym owners and staff**, who lose time searching for records and make mistakes in payments and renewals, and **members**, who face confusion about their subscription status. It is important because manual records lead to lost income, wrong data, and poor customer service.

3 Background

Gyms are a common part of daily life, and many of them operate as small, owner-run businesses. These gyms often rely on paper registers, notebooks, or basic spreadsheets to keep member details, subscription dates, and payments. Because this information is stored in different places, staff must check several records just to confirm the status of one member.

As the number of members grows, this approach becomes slow and error-prone. Applying software engineering practices such as requirements analysis, structured design, and systematic testing allows us to build a reliable and easy-to-use system that replaces these manual methods.

4 Target Users

Gym Administrator / Owner	Receptionist / Staff	Gym Members
Manages members, plans, staff, and reports.	Registers members, records payments, and checks attendance.	View their own subscription status and payment history.

5 Project Objectives

1. Store all member information in one organized and searchable database.
2. Manage membership plans (monthly, quarterly, yearly) and track subscription start and expiry dates.
3. Record payments and reduce errors in fees and renewals.
4. Record member check-in attendance at the gym.
5. Generate simple reports on members, income, and expiring subscriptions.

6 Proposed Solution

We propose a **desktop software system** with a simple graphical interface and a central database. Staff register members and assign a plan; the system calculates the expiry date automatically and records each payment.

It shows alerts for subscriptions that are about to expire or already expired, logs member check-ins, and produces summary reports. Different user roles (administrator and staff) will have different access levels.

7 Project Scope

INCLUDED IN THIS PROJECT

- ✓ Login system with user roles (administrator and staff)
- ✓ Member management: add, edit, search, and deactivate members
- ✓ Membership plan management and subscription renewal
- ✓ Payment recording with receipt details
- ✓ Attendance check-in log
- ✓ Reports: active and expired members, monthly income, and upcoming expiries

NOT INCLUDED (FUTURE WORK)

- ✗ Online payment
- ✗ Mobile app
- ✗ Biometric or card-reader hardware
- ✗ Workout or diet planning

8 Expected Benefits

Saves time
Replaces paper records and manual searching.

Fewer errors
Reduces mistakes in payments, renewals, and member data.

Better income tracking
Reduces missed or unpaid renewals.

Faster decisions
Quick access to accurate reports for the owner.

Better member experience
Faster service and clear subscription status.

9 Group Members and Responsibilities

No.	Student Name	Initial Responsibility
1	Moeen Rebar	Project leader; requirements analysis and documentation
2	Shady Jamil	System design (database and interface design)
3	Bilal Taha	Development, testing, and quality checks